

Armazéns de Dados

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Data Warehouse Architectures

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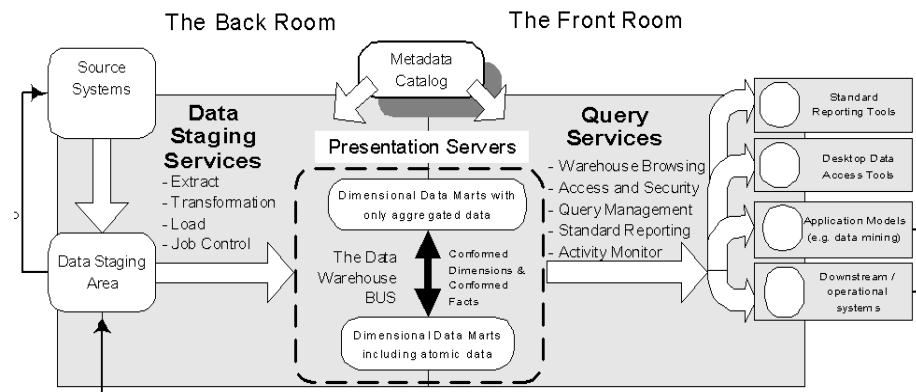
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Data Warehouse Bus Architecture (Ralph Kimball)

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DW BUS Architecture



- DW is built on a series of incremental data marts
 - **"Bottom-up"** or **incremental methodology**
- Has two major types of components: **services** and **data stores**

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DW BUS Architecture

- Is divided into two groups of components and processes
 - **Back-room (data acquisition)**
 - ♦ Part responsible for gathering and preparing the data
 - ♦ Where data acquisition and data staging processes take place
 - **Front-room (data access)**
 - ♦ Part responsible for delivering data to business users
- Flow of data from source systems to user desktop is supported by the **metadata catalog**
- Includes two types of data marts in the data presentation area
 - **Atomic data marts**
 - **Aggregated data marts**

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Back-Room

- Where the **data staging process** takes place
- **Engine room** of the DW
- **Primary concern:**
 - Getting the right data, with the appropriate transformations, at the right time and load it into the DW

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Source Systems

- Are the **obvious sources** of interesting business data
- Other high-value sources may be **external to the business**
 - Demographic customer information, target customer lists, and competitive sales data
- Data storage types are dictated by the source system
 - Many older legacy systems are standard mainframe data storage facilities: IMS, IDMS, VSAM, and DB2 are common
- **Flat file** is one often standard source for the DW
- Understanding their nature **is critical** for creating the back-room architecture

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Data Staging Area

- Is both a **storage area** and **set of processes** commonly referred as **Extraction, Transformation and Loading** (ETL), not seen by end-users
- **Everything between** the source system and the DW presentation server
- Where much of the **data transformation** takes place and much of the **added value** of the DW is created
 - **Cleaning the data**
 - ♦ Correcting misspellings, resolving domain conflicts, dealing with missing values, or parsing into standard formats
 - **Integrating data** from multiple sources
 - **De-duplicating data**
 - **Assigning surrogate DW keys**
 - ...

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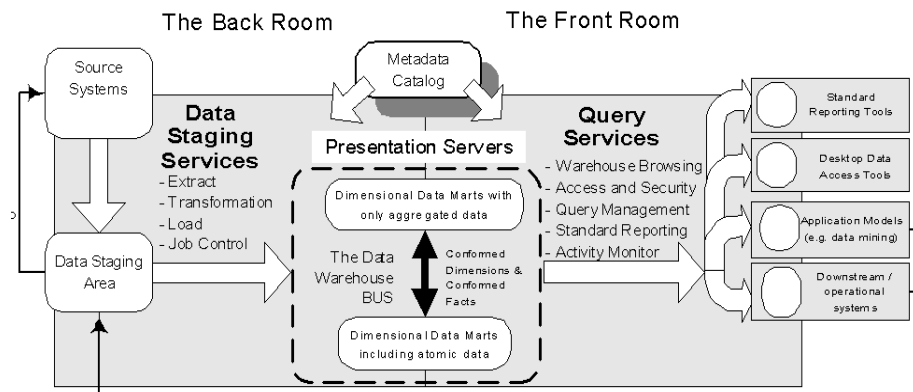
Back-Room Services

- **Tools and techniques** employed in the data staging process
 - Also known as ***data staging services***
- **Service** is an elementary function or task, that can be as simple as:
 - Creating a table in a database
 - Copying data from one table to another

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DW BUS Architecture



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Presentation Server

- Where the **data is stored** for direct querying by end-users, data analysis tools, reporting systems and other applications
- Is a **series of integrated data marts**
 - Data mart presents the data from a single business process
- Data in the queryable presentation server of the DW must be:
 - **Dimensional**
 - **Atomic** (to unpredictable ad-hoc user queries)
- All data marts must be built using **common/shared dimensions**

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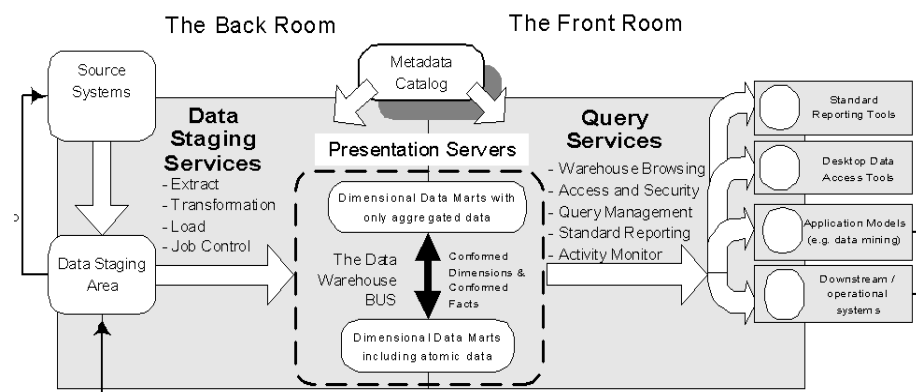
Data Marts in Presentation Server

- **Atomic data marts**
 - Hold multidimensional data at the **lowest detail level**
- **Aggregated data marts**
 - Hold multidimensional data that have been **aggregated**
 - Improve **query performance**
 - Loaded from the **data staging area** or from the **atomic data marts**
- All star schema-based data marts **may or not reside within the same database instance**
- Collection of star schemas which share dimensions and facts is the basis of the **DW Bus Architecture**

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DW BUS Architecture



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Front-Room

- **Public face of the DW**
 - It's what the business users see and work with day-to-day
- **Data access services are between the users and the data**, hiding some of the complexities and helping them to find what they are looking for

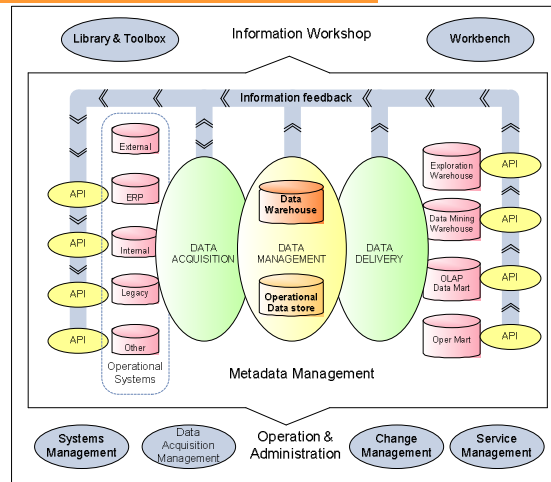
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Corporate Information Factory (Bill Inmon)

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CIF Architecture

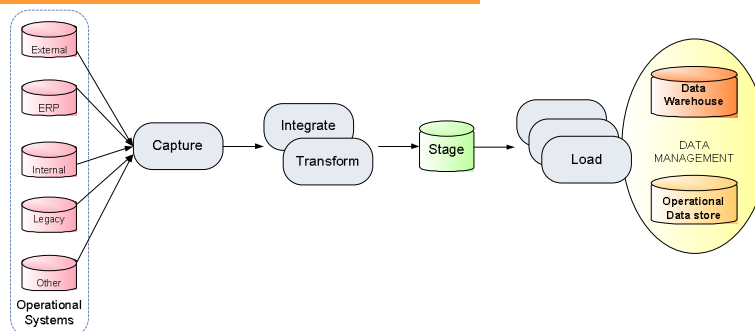


Logical architecture with **major databases** and **processes** to effectively and efficiently move data from source systems to business users

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Data Acquisition

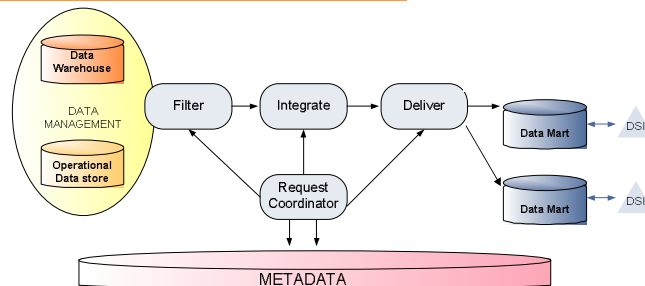


- Set of processes and programs that extract data from the operational systems to the **DW** and to the **Operational Data Store (ODS)**
- Perform the **cleaning, integration** and **transformation** of the data into an enterprise format

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Data Delivery



- Process that moves data from the DW or ODS into **data marts** and **oper marts**
- Like in the acquisition layer, data is manipulated as it is moved
- Origin is the **DW** or **ODS**, which already contains high quality integrated data that **conforms to the enterprise business rules**

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DW vs. ODS

- **DW** – A **subject-oriented, integrated, time variant** and **non-volatile** collection of data used in strategic decision making [Inmon and Hackathorn, 1994]
- **ODS**
 - Data is **fully integrated** like in a DW
 - **Data is current**
 - **Data is volatile or updatable** (no history is retained)
 - Data is **usually entirely detailed**
 - Source of **near real-time** and **accurate data** – accessible from anywhere in the corporation

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Data Marts vs. Oper Marts

- **Data marts** are customized and/or aggregated subsets of data derived from the DW
 - Where the analytical activities take place
- Data in each data mart is usually **tailored for specific analytical requirements** of a business unit or function
 - Product profitability analysis, sales analyses, ...
- **Oper marts** are derived from the **ODS** and used to provide the business community with **dimensional access to current operational data**

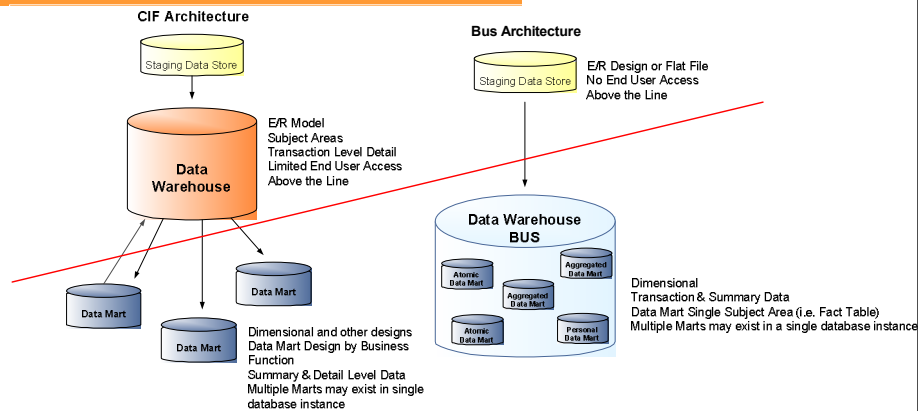
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Bus Architecture vs CIF Architecture

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Comparing CIF and BUS Architectures



Access is usually not allowed above the red line

- Back room is completely off-limits to users in the BUS architecture
- Direct access to the CIF DW is discouraged

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CIF vs. BUS: Differences

- **No physical repository equivalent** to the CIF DW in the BUS architecture
- BUS DW is the collection of **atomic and aggregate data marts**
- BUS architecture data marts (**star schemas**) are significantly different from the design of the CIF DW (**relational schema**)
- Various data marts schemas “conform” through **common dimensions** in the BUS architecture
- In the BUS architecture **all components are dimensional, except the data staging area**

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CIF vs. BUS: Similarities

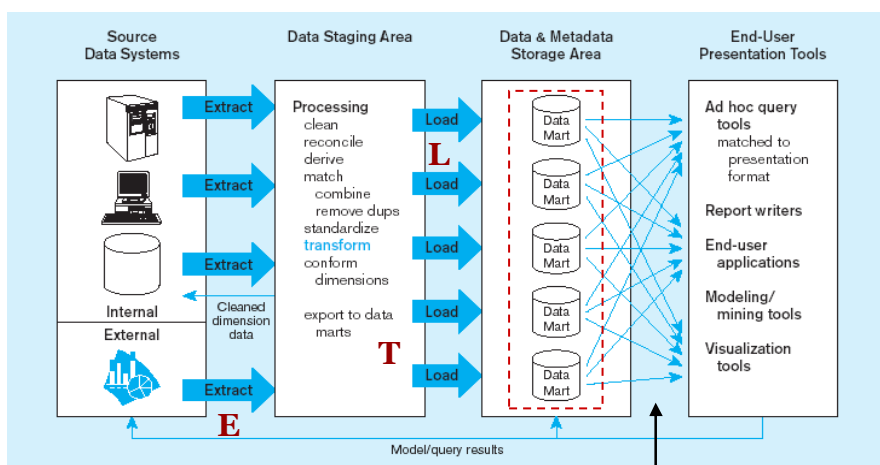
Both architectures:

- Have a **separate staging area, metadata management, data acquisition** and **data delivery** processes
- Power of information resides in the **atomic data**, which embed all available information dimensionality
- Existence of **dimensional data marts**
 - **Aggregate data mart** in BUS architecture is **usually the same** as the **data mart** in the CIF architecture

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Kimball DW Architecture



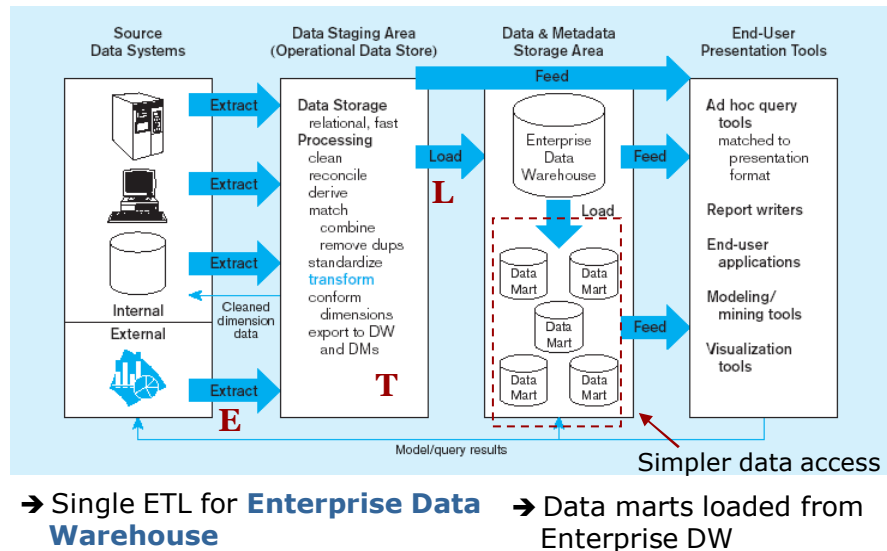
→ Separate ETL for each **independent** data mart

→ No single consolidated DW from the beginning

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Inmon DW Architecture



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Kimball Approach

- **Most common approach**
- Begins with **a single data mart** and **others are added over time** for more subject areas
 - Will require an overall **integration plan**
- Relatively **inexpensive** and **easy** to start to implement
 - Can be used as a **proof of concept** for DW
- Separate ETL process is developed for each data mart, which yields **costly redundant data and processing efforts**
- Can perpetuate the **"silos of information" problem**
- Key is to have an **overall plan for integrating the different data marts**

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Inmon Approach

- **Comprehensive DW is built initially**
- Data marts are built using **aggregate subsets of the data in the DW**
- Like all complex projects, it is **expensive, time consuming**, and **prone to failure**
- When successful, it results in an **integrated and scalable DW**

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